MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 829B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Push-pull, rf beam power amplifier

Outline	---	See figure 1
Base	---	See figure 1 and Group No. 6 (EIA)
Envelope	---	See figure 1
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	Top	leads	(see note 1)
Element	---	h	2g1	g2	k	hct	1g1	h	1a	2a	

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Ib	Ic
Unit:	V	Vdc	Vdc	Vdc	v	mAdc	mAdc
C Telep:							
Maximum (natural cooling):	6.9 or 13.9	600	-175	225	100	212	15
Maximum (forced-air cooling 1/):	6.9 or 13.9	600	-175	225	100	212	15
Minimum:	5.7 or 11.3	---	---	---	---	---	---
C Teleg:							
Maximum (natural cooling):	6.9 or 13.9	750	-175	225	100	240	15
Maximum (forced-air cooling 1/):	6.9 or 13.9	750	-175	225	100	240	15
Minimum:	5.7 or 11.3	---	---	---	---	---	---
TEST CONDITIONS:	6.3 or 12.6	400	-11	225	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Pp	Pi	Pg2	TE	Fi	Modulation	Alt
Unit:	W	W	W	°C	MHz	---	ft
C Telep:							
Maximum (natural cooling):	21	67.5	7	235	200	Anode	10,000
Maximum (forced-air cooling 1/):	28	90	7	235	200	Anode	10,000
Minimum:	---	---	---	---	---	---	---
C Teleg:							
Maximum (natural cooling):	30	90	7	265	200	---	10,000
Maximum (forced-air cooling 1/):	40	120	7	235	200	---	10,000
Minimum:	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---	---

GENERAL:

Qualification - Required
Holding period (72 hours)

1/ See note 2

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1226	Envelope temperature	Pp = 40 W static dissipation (note 6 except 1,600 ft minute maximum)	---	---	T	---	235	°C
	Quality conformance inspection, part 1							
1266	Total grid current	Ec/Ib = 50 mAdc; t = 30 seconds (see notes 3 and 4)	0.65	II	Ic1	0	-4.0	µAdc
1301	Heater current	Ef = 6.3 V	0.65	II	If	2.0	2.5	A
1256	Electrode current (screen)	Eb = 250 Vdc; Ec2 = 175 Vdc (see note 4)	0.65	II	Ic2	0	10.0	mAdc
1256	Electrode current (anode)	Eb = 250 Vdc; Ec2 = 175 Vdc (see note 4)	0.65	II	Ib	38	82	mAdc
1236	Power oscillation	Ebb = 400 Vdc; Ec2 = 225 Vdc maximum; Ib = 250 mAdc (total 2 anodes); Rg = 5,000 to 15,000 ohms per tube; Ic = 10 to 15 mAdc (total 2 grids); Ic2 = 35 mAdc maximum; F = 200 MHz; t = 30; push-pull (see note 3)	0.65	II	Po	45	---	W
1231	Emission	Eb = Ec1 = Ec2 = 25 Vdc (see notes 3 and 4)	0.65	II	Is	250	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1336	Heater-cathode leakage	Both filaments energized	---	---	Ihk	---	175	µAdc
1331	Direct-interelectrode capacitance	See note 5	---	---	Cga	---	0.12	pF
					Cin	12.8	16.2	pF
					Cout	5.2	8.7	pF
1036	Bump	Angle = 20°	---	---	---	---	---	---
1031	Low-frequency vibration	Eb = 250 Vdc; Ec/Ib = 10 mAdc; Rp = 2,000 ohms (see note 4)	---	---	Ep	---	800	mVac
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
---	Life test	Group C: Ebb = 750 Vdc; Ec2 = 225 Vdc (max); Ib = 160 mAdc (total 2 anodes); Rg = 7,500 ±500 ohms per tube; Ic = 7.5 to 15 mAdc (total 2 grids); F = 200 MHz (see note 6)	---	---	---	---
---	Life-test end points (500 hours)	Power oscillation Heater-cathode leakage	Po Ihk	37 ---	--- 175	W μAdc
(B) 1031	High-frequency vibration	No voltages applied; t = 900 seconds (see note 7)	---	---	---	---

NOTES:

1. Top lead over pin No. 2 is 2a; top lead over pin No. 6 is 1a.
2. Cooling shall be provided which is equivalent to that obtained in the physical layout shown on figure 2, with an air velocity of 2,000 ft/minute minimum as read on the 3,000-ft/minute scale of the specified instrument. The air requirements stated above are based on operation at sea level and an ambient temperature of 20° C.
3. This test to be performed at the conclusion of the holding period.
4. Read on both units separately. Control grid of unit not under test shall be connected to minus 100 volts.
5. Duplicate test on each unit separately. The unit not under test to ground. The C_{g_2} shall be measured with a shield 3/4-inch (19.05mm) high and 2-3/8 inch (60.33 mm) ID.
6. Forced-air cooling is required according to the physical layout shown on figure 2. An air velocity of 2,000 ft/minute maximum as read on the 3,000-ft/minute scale of the specified instrument shall be obtained. The minimum ambient temperature shall be 20° C.
7. (B) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Navy -
Air Force - 80
DSA - ES

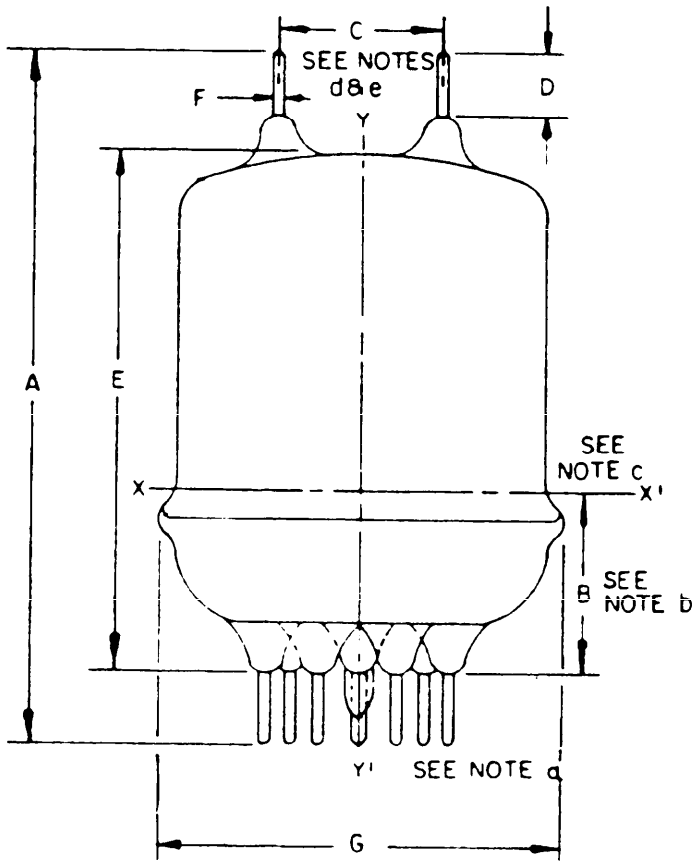
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	3.930 (99.82)	4.313 (109.55)
B	See note b	
C	See notes d and e	
(B) Quality conformance inspection, part 3 (see note 7)		
D	.375 (9.53)	.625 (15.88)
E	2.875 (73.03)	3.250 (82.55)
F	.052 (1.32)	.060 (1.52)
G	---	2.375 (60.33)

FIGURE 1. Outline drawing of electron tube type 829B.

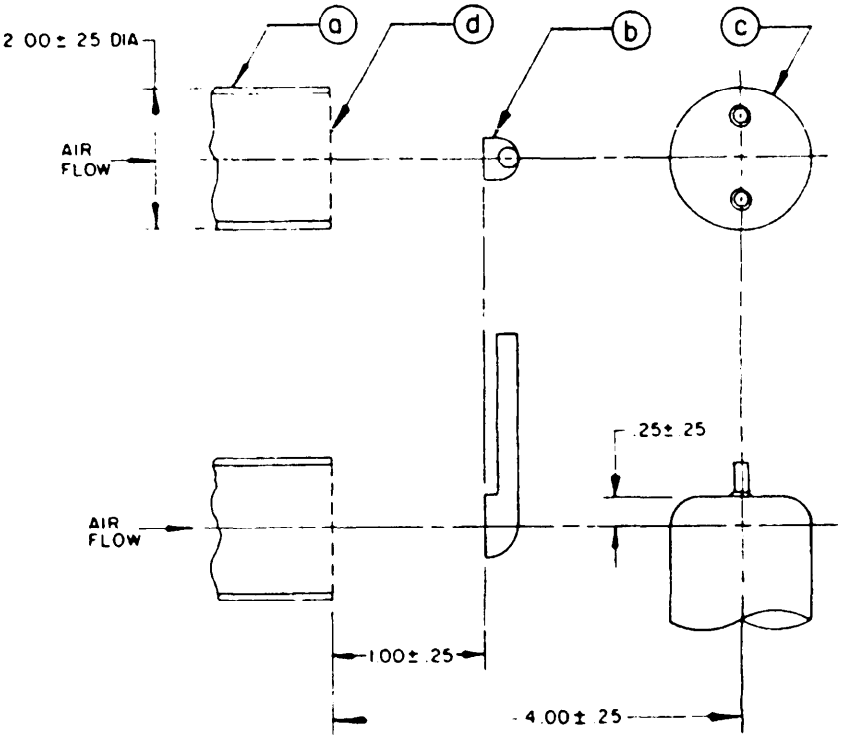
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NOTES:

a. The axis YY' is defined as the axis of the base pin gage. (EIA Group No. 6).
b. Dimension "B" is measured by inserting the tube in the base-pin gage and then lowering a gage plate having a hole 2.063 (52.40 mm) +.003 (.08 mm) -.000 (.00 mm) in diameter until the plate rests on the seal flange at position XX' . The centerline of the hole shall be coincident with the axis YY' within .150 (3.81 mm). With the gage plate parallel to the top surface of the base pin gage, the dimension "B" is measured between the bottom surface of the gage plate and the top surface ot the base pin gage. This distance shall be .844 (21.44 mm) min. and 1.219 (30.96 mm) max.
c. Minimum diameter of tube-seal flange will be such that a gage having ID of 2.125 (53.98 mm) +.003 (.08 mm) -.000 (.00 mm) and thickness of .125 (3.18 mm) ±.010 (.25 mm) will not pass the flange when tried at any angle.
d. The anode pins shall be capable of entering a flat plate gage of .375 (9.53 mm) min. thickness having 2 holes .120 (3.05 mm) ±.0005 (.01 mm) in diameter arranged .424 (10.77 mm) ±.001 (.03 mm) from a point coincident with the axis YY' . The axes of the holes shall be parallel to YY' and the plane of these axes shall be 90° ±5' to planes YY' and pin No. 4.
e. The anode pins shall be capable of entering a flat plate gage of .375 (9.53 mm) min. thickness having 2 holes .120 (3.05 mm) ±.0005 (.01 mm) in diameter arranged .848 ±.001 (21.54 ±.03 mm) (center to center).

FIGURE 1. Outline drawing of electron tube type 829B - Continued.

INCHES	MM
.25	6.35
1.00	25.40
2.00	50.80
4.00	101.60



ⓐ Air Duct.
ⓑ Alnor Velometer Model No. 3002 Jet No. 2220, or equal.
ⓒ Tube Type 829B under test.
ⓓ Mesh screening.
Note: Velometer shall be removed during the life test.

FIGURE 2. Forced-air cooling.

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